

Experimental pragmatics

1 Introduction

Pragmatics is a field of research that deals with language use in context and inference-making. At the center of it are questions related to how language users understand one another under uncertainty and in the face of linguistic ambiguity, how they close the gap between what is said in an utterance and what is meant by the utterer, and how these processes are grounded in human social interaction and cognition. As early as the 1980s, researchers started employing methods from experimental psychology to investigate these questions, which otherwise had been subject to fieldwork and small-scale qualitative analyses, or, in many cases, introspection and armchair speculation alone. This chapter deals with the experimental study of pragmatics, and the research field of the same name that has emerged and established itself in the last 30 to 40 years.

The chapter is structured as follows: Section 1 provides an overview of the development of experimental pragmatics as an empirical approach to the study of language use (1.1) and a discussion of how two classic topics in pragmatics have been investigated under the light of experimentation (1.2). Section 2 is focused on methodology and illustrates how experimental methods have been used to investigate real-time pragmatic processing (2.1) and how computational methods have been used to model experimental pragmatic data (2.2). Section 3 provides an overview of current research trends in the field, and Section 4 presents a short conclusion.

1.1 Establishment and development of the field

A quick overview of the literature shows that “experimental pragmatics”, as a term referring both to an approach within pragmatics and a research field of its own, has been growing in the last couple of decades: three separate handbooks feature chapters on experimental pragmatics (Huang, 2017; Rueschemeyer & Gareth Gaskell, 2018; Zufferey & Gigax, 2023), and not only is there a dedicated handbook of experimental

semantics and pragmatics (Cummins & Katsos, 2019) but also a number of collected volumes and monographs dedicated to the theme (Noveck & Sperber, 2004; Noveck, 2018, see also the Palgrave Studies in Pragmatics, Language, and Cognition monograph series). Historically, the movement started in psychology, where researchers working on inferencing and reasoning became increasingly more interested in the linguistic aspects of pragmatics and the interrelation between language and higher-order cognitive processes, in particular reasoning and what is known as theory of mind or perspective-taking (Premack & Woodruff, 1978). Right after the turn of the century, Noveck and Sperber (2004) edited the first volume on what was then starting to be known as experimental pragmatics, a book that has marked the beginning of the field as it's known today.

As stated by Sperber and Noveck (2004) in the introductory chapter to their edited volume, issues related to the study of human language use had thus far been investigated in pragmatics and psycholinguistics without much interaction between the two. Pragmaticians, on the one hand, relied heavily on intuitions about interpretation and small-scale observations of behavior. Psycholinguists, on the other hand, while employing quantitative and experimental methods to study human verbal behavior, were not using them either to test explicit pragmatic theories or to systematically tease apart issues of pragmatic interest. While the pioneering work of Noveck and others (see Noveck & Sperber, 2004, for an overview of early work) has helped bridge that gap, according to Sperber and Noveck (2004), a critical issue has not only been the disconnect between those working on pragmatics and those doing experimental research on human communication but rather the fragmentation of the field of pragmatics itself: there are at least two main strands of linguistic pragmatic thought, and a number of different theoretical traditions clustered within different frameworks, which makes communication and cross-fertilization between different corners of the field particularly hard.

As discussed by Huang (2009) in an influential textbook on pragmatics published just a few years after Noveck and Sperber (2004), research in the field can roughly be said to follow either the Anglo-American school of thought or the European Continental one, where the main difference between the two is the former's focus on cognitive and, in many cases, formalizable aspects of communication and inference-making and

the latter's focus on function and the situated nature of human language use. Crucially, the Anglo-American school of pragmatic thought is rooted in a linguistic and philosophical tradition which places pragmatics alongside other 'core' linguistic components like semantics, syntax, and phonology, assuming fairly sharp distinctions between them. In contrast, the European Continental school is rooted in a scholarly tradition which emphasizes the functional character of language use, where pragmatics is seen as a lens through which all aspects of grammar and human communicative behavior more generally can be explained. Experimental pragmatics has been built mostly on the Anglo-American tradition, incorporating and further developing many of the key assumptions and research topics of this tradition. Particularly relevant to the empirical work which has been carried out in the field is the division between so-called Neo-Gricean pragmatic theories and Post-Gricean ones. While both theoretical families build on the philosophical work of Paul Grice (e.g. Grice, 1975), they diverge when it comes to what is assumed to be a set of satisfactory explanatory principles behind the interpretive processes of ordinary language use. The next section presents a short review of two research topics which have received ample coverage both theoretically as well as empirically in experimental pragmatic research. In each case, the discussion is focused on how experimental pragmatic work has shed light on theoretical questions, and how it has informed subsequent theorizing. Section 2 then delves into details about the setup of classic experimental studies dealing with the topics discussed below.

1.2 Investigating classic topics experimentally

1.2.1 Implicature derivation

One of the key topics, if not the most well-represented topic, in the Anglo-American pragmatic research tradition is the derivation of implicatures. While there is much discussion about what exact typology of interpretations best characterizes the interpretive processes at hand in human communication (Ariel, 2016), there is consensus in the field that implicatures are an essential part of it. By implicature derivation is usually understood an inferential process whereby a language user infers meaning from a linguistic expression where that same meaning is not made explicit, as in (1) below.

(1) (from Noveck & Sperber, 2004)

Peter: Do you like Fellini's films?

Mary: Some of them.

In the above example, Peter may infer that Mary means “some but not all of Fellini's films”. Crucially, that is different from what is literally stated in Mary's utterance, and the closing of the gap between the two is what is called the derivation of an implicature, the implicature being the intended, or inferred, meaning. This inference is thought to be guided by a combination of assumptions which may involve the strength of claims but also cooperative and trustworthy language use. While different theoretical accounts of the phenomenon make a different number of assumptions (see Nys, Wong, Schaecken, 2023), key is that language users are assumed to prefer strong claims over weak ones, which means that the use of a weaker expression like *some* may implicate that a stronger one like *all* does not apply. In this case, *all* is considered to be stronger than *some* as it makes a stronger claim on semantic grounds, meaning that it is true in more cases than its counterpart. The two terms, called scalars as they form an ordered scale where *all* is higher up informationally than *some*, stand in relation to one another and it is exactly based on this relation that interpreters may arrive at pragmatically-enriched interpretations involving the narrowing of the meaning of *some*.

The derivation of implicatures, and of scalar implicatures as in (1) above in particular, has received plenty of attention in the experimental pragmatic literature (see Bambini & Domaneschi, 2024, and the papers therein). Experimental work has been conducted on a wide variety of lexical scales, which are usually denoted by $\langle \text{term}_1, \text{term}_2, \text{term}_3, \dots \rangle$ and range from quantificational scales like $\langle \text{some}, \text{all} \rangle$, as in the example above, to adjectival (e.g. $\langle \text{possible}, \text{certain} \rangle$, $\langle \text{warm}, \text{hot} \rangle$) and verbal ones (e.g. $\langle \text{like}, \text{love} \rangle$). A key theoretical debate which experimental work has started shedding light on is that concerning the steps in the derivation of scalar implicatures. There are two families of accounts, which are usually pitted against one another: default and literal-first/ contextualist accounts. Default accounts assume that implicatures are derived automatically and effortlessly, while literal-first/ contextualist accounts assume that they are the result of a pragmatic process which is based on reasoning about an utterer's intentions and epistemic state in context. Crucially, these differences in the elements or steps involved in the inferential process are assumed to correlate with

differences in how comprehenders process and interpret utterances which may give rise to implicatures. For instance, default accounts predict no difficulty in deriving an implicature, whereas literal-first/ contextualist accounts, despite differences between them, predict some sort of difficulty in the derivation process compared to interpreting content which is not implicated.

Experimental pragmatic research in the last two-odd decades has largely accrued evidence for the literal-first/ contextualist view, suggesting that scalar implicatures do not arise by default but instead involve effort and some sort of processing cost, which is usually indexed by longer or delayed behavioral responses, extended neural and physiological processing signatures, or lower accuracy rates in offline interpretation tasks. However, as discussed in detail in recent reviews of the phenomenon (Khorsheed & Gotzner, 2023; Khorsheed, Price, & van Tiel, 2022), the source of the well attested processing costs is still a matter of much debate. According to Khorsheed et al. (2022), three main contenders are discussed in the literature: the informational or semantic complexity of an utterance, the actual disambiguation between a plain and a pragmatically-enriched interpretation, and a number of different yet related issues to do with the interpretation of negation and negated expressions.

Key findings to have emerged from the experimental literature are (a) that there seems to be an inferential cost in deriving a scalar implicature independent of the semantic complexity of the utterance (Bott, Bailey, & Grodner 2012), (b) that the decision to derive an implicature but not the derivation process per se might be what is costly (Marty & Chemla, 2013), and (c) that scalar inferences involving negation and negative polarity tend to be computationally more complex than their non-negated/ positive counterparts (Agmon, Loewenstein, & Grodzinsky, 2019; Bott, Schlotterbeck, & Klein, 2019; cf. Schwarz, Romoli, & Bill, 2015).

Further attesting to the role experimentation plays in the advancement of scalar implicature research, Khorsheed and Gotzner (2023) highlight that variability in the experimental results found in the literature has to do not only with inter-individual variability related to internal cognitive factors and external social factors but also with the particular method employed in a study and the language investigated. All in all, this stresses even more the need for research using diverse and complementary

methods (as discussed in some more detail in section 2), and for comparing scalar phenomena not only within but also across languages. Diversity in the linguistic structures which give rise to scalar implicatures is well attested, and there is growing debate as to the extent to which their associated inferences can really be unified under a single theoretical notion (Orr, Ariel, & Shetreet, 2024).

In short, experimental work has been key to advancing our theoretical understanding of the derivation of scalar implicatures, and it continues to be paramount to shaping this line of pragmatic research. Avenues for future research involve not only cross-referencing between different experimental methods and verifying derivation rates and processing signatures but also further examining the extent to which the theoretical notion of scalar implicature, with all of its variability, corresponds to identifiable cognitive processes (Katsos & Kissine, 2025).

1.2.2 Figurative language understanding

Another key topic in the Anglo-American pragmatic tradition which has received plenty of attention in the experimental literature is the understanding of figurative or non-literal language. While the topic itself stretches much beyond the boundaries of pragmatics, pragmatic research has been key to its theoretical and empirical development. A classic and well-studied example of figurativity in language are metaphors: non-literal statements which involve comparisons between two unrelated conceptual domains, such as in *life is a journey*, where the source domain of ‘traveling’ or ‘journeying’ is compared to the target domain of ‘life’ (Lakoff & Johnson, 1980), or the *weary mountain*, where the comparison is between the source domain of ‘weariness’ and the target domain of ‘mountains’.

Much like the case of implicatures, there is intense debate in metaphor research as to what the exact processes or steps involved in the interpretation of metaphoric utterances are (see Gibbs & Colston, 2012, and Holyoak & Stamenkovic, 2018, for reviews of the literature). Psychological accounts focus on cognitive processes and assign a primary role to either analogy or categorization. Analogy accounts assume that there is the computation of relational similarities between the two elements in the comparison (see Genter & Clement, 1988; Bowdle & Gentner, 1999, 2005), whereas categorization accounts suggest that there is a direct comparison between the elements

themselves (see Glucksberg & Keysar, 1990)¹. Crucially, from a processing perspective, the main difference is that analogy accounts posit a comparison across multiple elements within each domain, which is then followed by an abstraction process, whereas categorization accounts posit the comparison of a single element between the source and target domain, which follows an initial abstraction process. This implies, at least in principle, a higher processing cost for analogy, based on the fact that it assumes a comparison between multiple features as opposed to a single one as in the case of categorization. Aside from accounts rooted in psychology, there are a number of accounts stemming mainly from pragmatics which put more of an emphasis on the relation between literal and non-literal aspects of metaphors (e.g. Carston, 2010; Gibbs, 1994; Giora, 1997; Sperber & Wilson, 1986; Wilson & Carston, 2007). They generally predict that metaphors are as easy to process as non-metaphoric language provided that the context of utterance supports a pragmatic, non-literal interpretation. While agreeing on the facilitatory role of the larger discourse context, these accounts diverge from one another when it comes to the role they assign to the literal meaning of the metaphor terms. So-called direct access accounts assume that metaphors can be interpreted without activation of their literal meaning (Gibbs, 1994; Sperber & Wilson, 1986), whereas other pragmatic accounts assume that the literal meaning, while potentially not disruptive, is still activated in processing (Carston, 2010; Giora, 1997; Wilson & Carston, 2007). In principle, a higher processing cost is predicted for the latter, though activation of the literal meaning is not per se equated with processing difficulty.

One key issue in early work on metaphor understanding was whether literal and metaphoric statements are processed similarly. While this position was empirically rejected relatively early on (see Gibbs, 1994, for an overview of early empirical work), work on the role of context in the facilitation of metaphor processing has continued. Generally, it has been established that metaphors are harder to process than literal language though context can attenuate that effect. Importantly, not always is it the case that differences between metaphoric and literal statements are indexed by

¹See also conceptual mapping accounts as put forward, e.g., by Lakoff & Johnson (1980) and Lakoff & Turner (1989), which, despite being related to categorization accounts, arguably form a category of their own as they do not necessarily make processing predictions.

processing times per se. Rather, differences may emerge in offline interpretation behavior or other processing signatures like neural activity as measured via event-related potential (e.g. Bambini et al., 2018; Pynte et al., 1996). Two important dimensions of variation which have been shown to impact metaphor processing are aptness and conventionality. Metaphor aptness relates to how salient properties of the source express important features of the target. For instance, in “Dancers can be butterflies” the source term ‘butterflies’ is a more apt descriptor of the target ‘dancers’ compared to “Soccer players can be butterflies” (Jones & Estes, 2006), given that the salient property attributed to the target, namely something akin to ‘graceful’, is more typical of the former than of the latter. Conventionality, on the other hand, relates to how established a metaphor is, where any given mapping can go from completely novel to highly conventionalized and even as far as not being figurative anymore. For instance, “This surgeon is a butcher” consists of a conventionalized mapping where not only low-level similarities between the source (butchery) and target (surgery) domains are evoked (e.g. butcher and surgeon, animal and human, commodity and patient) but also a sense that the referred surgeon is being evaluated as incompetent (Grady, Oakley, Coulson, 1999). As argued by Bowdle and Gentner (2005) in a reinterpretation of their early ideas, this so-called career of metaphor brings categorization and analogy closer together, as novel metaphors are claimed to be processed analogically, whereas more established ones are claimed to be processed categorically, as they lexicalize over time.

Put together, the findings of decades of experimental work on metaphor understanding seem to suggest mostly that some form of categorization process is at play, though the empirical picture is even more convoluted than in the domain of implicatures. For simple, nominal metaphors like *life is a journey*, the source term (i.e. ‘journey’) shifts towards a more abstract categorical meaning (e.g. a process, a development). However, the situation is less clear for structurally complex metaphors, such as proportional metaphors like *religion is the opium of the people*, or extended metaphors which repeatedly draw from the same source domain (e.g. talking about a debate repeatedly using terms related to racing, Gentner et al., 2001). It is also the case that while there may be differences in how novel and conventionalized metaphors are processed, in some studies aptness has been found to be a better predictor of metaphor comprehension than conventionality (Jones & Estes, 2005,

2006). All in all, semantic properties associated with the actual lexical forms used in a metaphor seem to matter perhaps as much as the cognitive processes involved in comparing conceptual domains, in line with pragmatic accounts of metaphor interpretation that assume some involvement of the literal meaning.

Avenues for future research involve extending the diversity of languages investigated, which is particularly narrow in the experimental literature on metaphor understanding, as well as examining more thoroughly what the exact role of context is, particularly in complex usages in which metaphors are extended and distributed but also where the mappings are not as clear-cut as in the case of simple nominal metaphors (Gibbs, 2023).

As in the case of implicatures, future work should aim to identify what exact cognitive processes are involved in the understanding of different types of metaphor, and whether or not they are similar to those involved in other instances of figurative and creative usages of language. On the one hand, experimental work has shown that a general theory of metaphor understanding will need to reconcile not only the roles of analogy and categorization but also those of psychological processes and the linguistic properties of the metaphor terms. On the other hand, as discussed with regard to other forms of pragmatic interpretation, the role of any single interpretive process may need to be reconsidered in more nuanced and less categorical ways, taking richer contexts and individual variability more into account.

2 Methodology

As suggested so far, methodological plurality and diversity are a cornerstone of experimental pragmatics. This section exemplifies how experimental methods have been employed in classic real-time processing studies in the domains of scalar implicature derivation and metaphor understanding, and it discusses the application of computational models to experimental data from those same domains.

2.1 Real-time processing

Work on language comprehension has formed the backbone of experimental pragmatics since the field was established. As discussed for the cases of scalar

implicatures and metaphor comprehension, much of this work tends to look at online processing: how comprehenders process and interpret language in real time as an utterance unfolds. The methods employed in this line of research have spanned classic psycholinguistic methods, neurolinguistic methods, and other, less common methods from experimental cognitive psychology. Psycholinguistic methods involve measuring reaction or reading times, other reading behavior like regressions, and gaze behavior relative to visually presented stimuli. Neurolinguistic methods involve measuring event-related potentials (ERPs) in electroencephalography (EEG) and using functional neuroimaging (fMRI) (see Bambini, 2010, and Gutierrez Cisneros, Foucart, & Brunelliere, 2026 for reviews of neuropragmatics). Other methods which have more seldomly been used are mouse-tracking and physiological measures such as pupil dilation, facial muscle activation (fEMG), and grip force.

In an online processing setup, measurements are locked to particular linguistic stimuli or utterance segments, which can be presented to participants either as text or auditorily. The goal is to measure a processing signature after the onset of the critical stimulus which is of theoretical significance. This can be used to compare signatures across experimental conditions, looking for indices of facilitation/ processing difficulty, which, as discussed earlier, can indicate a number of different things theoretically. In studies looking more specifically at incremental interpretation and predictive processing, a window of interest is established usually between the offset of the pragmatically relevant stimulus and the onset of a subsequent stimulus in the linguistic signal.

In scalar implicature research, the critical stimuli tend to be the scalar terms themselves, or some other linguistic expression in the utterance which is expected to be more or less felicitous given the use of a particular scalar expression. Bott and Noveck (2004), for instance, presented participants with sentences like “Some elephants are mammals” and measured the time participants took to indicate whether the statements were true or false, starting at the offset of the sentence. In other words, in their setup the scalar term right at the beginning of the sentence (originally presented in French) was the critical stimulus. Across a series of experiments, they found that participants took longer reacting to ‘some’ sentences when interpreting them pragmatically compared to sentences containing literal interpretations of ‘some’

and 'all'. Bott and Noveck interpreted this finding as evidence that enriched interpretations are not derived by default but rather depend on interpreters deciding to render an utterance more informative. Huang and Snedeker (2009), on the other hand, asked participants to look at visual displays containing pictures of children and objects and to "point to the girl that has some of the socks" while recording their eye movements. In this case, the critical stimulus was also the scalar term, and a window of interest was established between the onset of the scalar and before the completion of the phrase containing the target noun 'socks'. The measure of interest was the proportion of looks towards the picture compatible with a scalar interpretation of 'some' within that window. Crucially, the visual display included a target picture depicting a girl with two socks, which was compatible with the scalar interpretation, and a competitor picture depicting a girl with three soccer balls, which was incompatible with the scalar interpretation given that it contained all balls depicted in the display. The display also included a distractor picture depicting a boy with two socks and another irrelevant distractor picture (a boy with no objects). Huang and Snedeker found that on 'some' trials people took longer to fixate on the target compared to trials with other quantity expressions which could only be interpreted semantically (namely, 'all', 'two', and 'three'), though in all cases participants did so before the sentence was disambiguated at the end of the noun phrase. They interpreted this as evidence that a pragmatic interpretation of 'some', while available early, is nevertheless more difficult than plain, un-enriched interpretations of other salient scalar alternatives.

In metaphor comprehension research, the critical stimuli tend to be expressions which make a sentence metaphorical. Such expressions tend to be either nominal in nature, as in *life is a journey*, or adjectival, as in *weary mountain*. Glucksberg, Gildea, and Bookin (1982), for instance, presented participants with sentences like "Some jobs are jails" and measured the time they took to indicate whether the statements were literally true or not, starting at the onset of the sentence. In their setup the critical stimulus was the noun 'jails'. Glucksberg and colleagues found that people took longer responding to the metaphorical sentences, which were literally false, compared to literal sentences which were also false (e.g. "Some apples are oranges"). They interpreted this as evidence that people interpret both the literal and non-literal meaning of a sentence as they process them. Gentner and Imai (1992) presented

participants with sentences like “Christmas is six days before New Year’s Day” and measured the time they took to indicate whether the first event in the sentence took place either in the past or future relative to the second event. In this case, while the critical stimulus is still the entire sentence, the expression making the sentence metaphorical is neither a noun nor an adjective but rather the adverb ‘before’, which establishes a parallelism between the temporal relation in the sentence and contiguity in the physical sense. Genter and Imai presented the critical sentences preceded by setting sentences which established a space-time parallelism that was either ego-moving, where front is assigned to the future (later in time), or time-moving, where front is assigned to the past (earlier in time). They manipulated the mapping between the critical and setting sentences, such that there were consistent mappings (i.e. both sentences used the same space-time mapping) and inconsistent ones (i.e. one sentence used the ego-moving mapping and the other the time-moving one). They found that people took longer to respond when the test sentences formed inconsistent mappings with the setting sentences (i.e. ego-moving with time-moving, and vice versa). Gentner and Imai interpreted this finding as evidence that a consistent global mapping between a metaphor and its larger discourse context facilitates processing.

In summary, classic studies in experimental pragmatics have relied on reaction time and visual world eye-tracking paradigms to investigate real-time pragmatic processing. Crucial for experimental pragmatic research focused on real-time comprehension are not only the critical stimuli themselves but also the timing of the measurements relative to those stimuli. As illustrated by the examples discussed above, empirically attested differences in processing signatures can index, for instance, difficulty in deriving scalar implicatures compared to interpreting un-enriched sentences containing scalars, or difficulty in understanding metaphors compared to comparable literal statements, both of which are informative for theoretical accounts of pragmatic interpretation.

2.2 Computational modeling

While computational models have long been used in psycholinguistic research as tools to extend and complement traditional research relying primarily on experimentation (Crocker, 1996), recently they have also started to be brought to bear on experimental work in pragmatics. In general, the goal behind the use of

computational modeling in experimental research is to generate quantitative predictions which can be validated using experimental data, whatever its exact nature. As discussed in more detail below, modeling pragmatic data involves isolating processes tied to either production or comprehension, or integrating production and comprehension data, which is particularly relevant for pragmatic phenomena.

In scalar implicature research, so-called Rational Speech Act (RSA) models have been widely adopted as a way of modeling scalar reasoning. In a nutshell, the RSA framework provides a flexible and hierarchical way of approximating both speaker and listener behavior by calculating probabilities over world states and utterances (Frank & Goodman, 2012; Goodman & Frank, 2016; see also Degen, 2022). Ultimately, what such a model outputs are probabilistic predictions for the various components of the model (Franke and Jäger, 2016, see section 3.1 for a more detailed overview of probabilistic approaches in pragmatics), which consist of production and interpretation protocols based on explicitly defined mappings between utterances and world states. In its simplest form, there are three basic protocols in an RSA model: one for the behavior of literal listeners who interpret utterances in line with the literal semantics defined in the model, one for pragmatic speakers who produce utterances in line with what they think literal listeners are most likely to correctly interpret in context, and finally one for pragmatic listeners who reason about what the most likely intended meaning behind utterances produced by pragmatic speakers might be. Model predictions can then be used to draw inferences about the underlying causal mechanisms involved in pragmatic reasoning, based on the specifics of the model architecture, such as how refined literal listeners are assumed to be.

Franke and Bergen (2020), for instance, have argued that computational statistical models in the spirit of RSA can be used in a theory-driven way to investigate pragmatic reasoning. They demonstrated how model comparison is useful in establishing the best model architecture based on empirical data, or in other words, what the most likely underlying causal explanation for the data generation process might be. Franke and Bergen modeled scalar implicature derivation using four model variants, a basic instantiation of the original Frank and Goodman (2012) RSA model where listeners are sensitive only to the un-enriched lexical semantics of ‘some’, a model where listeners are sensitive to potential uncertainty around the lexical

meaning speakers assign to ‘some’, where some speakers might have a literal and others an enriched lexical entry for the quantifier, a model where listeners are sensitive to the fact that speakers may use an ambiguous term like ‘some’ differently in every usage, and finally a model where listeners are aware of all the logically possible interpretations of ‘some’, including in complex nested sentences like “Some of the people drank some of the water”, where each ‘some’ might have a different interpretation. When fitting the four model variants to empirical data on human production and interpretation of sentences like “Some of the people drank some of the water”, the model with the best fit was the one where users are aware of the full range of semantic ambiguities provided by their language, such that they use them flexibly when producing and interpreting language.

In other work drawing on RSA modeling, Pelouquin and Frank (2016) explored the role of alternative sets in scalar implicature derivation. They inputted empirically elicited measures of both speaker and listener judgments about sets of alternatives to a model computing pragmatic judgments. Speaker judgments involved eliciting plausible alternatives to a target scalar, while listener judgments involved tapping into people’s perception of the compatibility of different scalar terms with a 5-point quantitative rating scale, i.e., people were shown a value judgment on the 5-point scale and asked whether it matched the target scalar, as in “Do you think the person liked the food”?, where ‘liked’ was the target. People were presented with different sets of alternatives, either a minimal set with the two terms needed to generate the relevant comparison of interest (i.e., ‘some’ and ‘all’, or ‘liked’ and ‘loved’), or sets with 4 (e.g. ‘all’, ‘most’, ‘some’, ‘none’) and 5 alternatives (e.g. ‘all’, ‘most’, ‘some’, ‘little’, ‘none’).

The authors then elicited pragmatic judgments to do with the use of scalars, where the compatibility task was reversed, meaning that people were shown a sentence with the target scalar (“Someone said they liked the food”) and asked to rate the respective value judgment using the 5-point scale. Crucially, this served to show that listeners draw inferences when making judgments based on the scalars, such that they were more likely to assign a higher rating when stronger terms were used compared to weaker ones, despite both strong and weak terms being, in principle, compatible with high ratings, as the first set of listener judgments shows. All elicited judgments were

then inputted into an RSA model, each set of judgments being mapped onto a different component of the model, namely literal speaker intuitions, literal listener intuitions, and pragmatic listener intuitions. The model estimated pragmatic behavior as indexed by judgment of utterances produced to describe the finite set of meanings related to the setup of the experiments (restaurant reviews), where a pragmatic listener attempts to reverse-engineer the most likely intended meaning given a target utterance. Model predictions were a better fit to human data when there was a wider availability of alternatives, in particular not only the two entailment terms (e.g. ‘some’ and ‘all’) but also two negative alternatives (‘most’ and ‘none’). What this shows is that the basic set of entailment terms, while minimally necessary for implicature derivation, may not be sufficient in explaining actual derivation rates.

Other modeling frameworks and types of models have also been used in experimental pragmatic research focused on scalar implicature derivation. Lately, transformers and Large Language Models have been at the center of attention. Hu, Levy, and Schuster (2022), for instance, have used a text-to-text transformer model (of the larger family of sequence-to-sequence models) to investigate scalar implicature derivation based on interpreters’ confidence in the underlying scale. More concretely, they used the model to estimate uncertainty over the distribution of possible alternatives as conditional on the context. They found that the model had a good fit to the experimental data reported in van Tiel, van Miltenburg, Zevakhina, and Geurts (2016), meaning that it predicted human data when it comes to predictability of strong scalemates, where that is not a good predictor of implicature derivation, as well as scale uncertainty, where lower uncertainty over the scalemates in a scale is associated with higher rates of implicature derivation.

In the domain of metaphor understanding, computational models have long been used for modeling metaphoric mappings as found in naturalistic text. In fact, computational and statistical processing of metaphoric language is a very prolific endeavor, with much attention dedicated to it in natural language processing research (see Shutova, Teufel, & Korhonen, 2013, and Rana, Chikkara, & Sharma, 2024 for overviews). Less commonly, however, have computational models been used to model human data as elicited experimentally. The models which have been used tend to fall within the remit of what is known as distributional or vector space semantics, an approach which

treats lexical meanings as high-dimensional vectors where the similarity between any two lexical items depends on the similarity between their respective vectors. Semantic spaces are said to be distributional as they are constructed on the basis of distributional occurrences of words in text. In early work, Kintsch (2000) demonstrated that word vectors can be used to model metaphor comprehension, namely by computing vectors of semantically-related words centered around the source and the target, and then computing the centroid between both of those vectors. The centroid is taken to reflect an *ad hoc* category of which the target is a member. Crucially, Kintsch and Bowles (2002) showed that vectors centered around both the source and the target (metaphor vectors), are better predictors of human literal paraphrases of nominal metaphors (e.g. *My lawyer is a shark* ~ *My lawyer is vicious*) than vectors centered around either only the source or only the target.

In subsequent work, Utsumi (2011) used word vectors to model both categorization and comparison processes, building on the early work by Kintsch (2000) focused on categorization. Utsumi (2011) modeled comparison via a neighborhood search that is conducted on both the source and the target simultaneously, unlike in the case of categorization, where neighbors for sources and targets are searched separately. After an initial number of common neighbors is found, a centroid is computed between that neighborhood and the target vector alone. Utsumi (2011) demonstrated that the comparison procedure is better at predicting metaphors with low interpretive diversity (i.e., where humans generate few interpretations and there is a single dominant interpretation, as in “One’s character is a stone”), whereas the categorization procedure is better at predicting metaphors with high interpretive diversity (i.e., where humans generate many interpretations with similar frequency, as in “An eye is a mirror”).

While these applications of vector space semantics shine a light on the potential underlying interpretive processes involved in metaphor understanding, a limitation of their usage in this domain is that the data which has been modeled was elicited using rather simple interpretation tasks, such as listing metaphor features (Kintsch & Bowles, 2002) or completing sentences with a literal paraphrase of a target metaphor (Utsumi, 2011). In other words, experimental pragmatic work investigating metaphor understanding is still to benefit from adopting computational modeling more widely,

as is the case in scalar implicature research. All in all, the work described in this section illustrates that computational models are powerful tools in the toolkit of experimental pragmatic researchers, and that the field has much to gain by integrating them more closely with the full breadth of empirical work being done experimentally.

3 Current trends

In this section, some of the current research trends in the field of experimental pragmatics are discussed. Given that the field has grown to be quite diverse both in its empirical and theoretical scope, here the focus is on work which is relatively underrepresented compared to major topics like implicature derivation and metaphor understanding, or work which expands classic research questions in novel and theoretically interesting ways.

3.1 Probabilistic pragmatics

As discussed in the section on methodology (2.2), probabilistic computational models have been used to model experimental pragmatic data with important implications for pragmatic theory. This work is part of a larger effort to bring probabilistic and information theoretic approaches to the study of linguistic and pragmatic phenomena (Bod, Hay, Jannedy, 2003; Gibson et al., 2019). As discussed by Franke and Jäger (2016), a probabilistic approach to pragmatics involves making general assumptions about the causal processes involved in language use based on tools of probability theory and a rationalistic perspective on cognition and human behavior. Despite being relatively general, these assumptions allow making more explicit connections between empirical data and their potential causal mechanisms, which in turn allows comparing empirical measurements of any sort to quantitative predictions generated by models fit for purpose.

Key to the probabilistic definition is the fact that such approaches assume inherent uncertainty in any pragmatic process, and that such uncertainty can be modeled using probability distributions. Crucially, these distributions are used both to represent language users' subjective uncertainty over world states and to describe interpretation as an abductive (Bayesian) inference over the most likely state associated with a linguistic utterance in context.

One particularly prolific endeavor in probabilistic pragmatics is the application of probabilistic models implemented computationally to experimental pragmatic data, as discussed in section 2.2 in the context of scalar implicature research. Aside from implicature derivation, another key topic which has received attention in this line of work is figurative language understanding. Here, Rational Speech Act models have been used to model the hyperbolic usage of numerical expressions such as “The electric kettle costs 1000 dollars” (Kao et al., 2014), and irony in evaluative statements such as “The weather is amazing” in reference to a blizzard (Kao & Goodman, 2015). In both cases, non-literality has been modeled as a function of affect, more concretely users’ intention to express their positive or negative affective stance on the communicated content (either the price of a home appliance or the weather, respectively). More recently, RSA models have been extended to take a more naturalistic view on the possible rhetorical goals behind figurative language use (Spinoso-Di Piano et al., 2025). This work has shown that a model formalizing rhetorical strategies, as opposed to affect-based reasoning, captures human judgment as well as or even better than affect-based models. In the case of hyperbole, the model described in Spinoso-Di Piano et al. (2025) distinguished between four possible rhetorical strategies, including not only literal and hyperbolic usages of numerical expressions (e.g. “The electric kettle costs 1000 dollars” meaning “it costs exactly 1000 dollars” or “it costs much more than expected”) but also understatements and rounding. In the case of irony, the model included both literal and ironic statements (e.g. “The weather is amazing” in the context of a blizzard meaning either ‘amazing’ or ‘terrible’).

While the RSA framework has become a popular modeling approach within probabilistic pragmatics, probabilistic pragmatic approaches also include other game theoretic and decision theoretic models (Benz & Stevens, 2018; Franke, 2013) as well as various other Bayesian models which also propose computational-level explanations for production and/ or comprehension behavior (see references below). Game and decision theoretic models have not often been used to model empirical data, with the notable exception of RSA models. Bayesian accounts of pragmatic phenomena, on the other hand, have usually been developed in order to account for psycholinguistic data. Some examples of the application of these models include

accounts of pronoun resolution and informativity, which have been used to model the production-interpretation asymmetry empirically attested in pronoun research (Kehler & Rohde, 2019), and predictability effects both in online language comprehension (Rohde, Futrell, & Lucas, 2021) and offline interpretation (Rohde et al., 2022).

All in all, probabilistic approaches have become an important component of modern experimental pragmatic work. Future work in this domain should strive to assess the ecological validity of the data probabilistic models are fitted to. Indeed, despite being powerful tools, probabilistic models of language are becoming increasingly more complex and focused on behavior often detached from the ways humans actually use language to communicate. As such, criticism has started mounting not only regarding their cognitive plausibility (Kwisthout & van Rooij, 2020; Woensdregt et al., 2024) but also their explanatory potential in light of social-interactive constraints (van de Braak et al., 2024). In this regard, future work should also aim to more explicitly test what the explanatory value of different probabilistic modeling approaches is, in particular when considering their relation to actual mechanistic cognitive processes.

3.2 Multimodal pragmatics

The analysis of multimodal human communicative behavior is by no means an emerging trend in linguistics and cognitive science of language (see Cienki, 2002; Holler & Levinson, 2019; Zima & Bergs, 2017). Indeed, questions related to the integration between gesture and speech or the interaction between different semiotic modes in text-based communication have long been studied in pragmatics and adjacent fields using corpus and computational approaches (see Allwood, 2008; Bateman, 2014; Huang, 2021, along with references above). Yet, despite also being prevalent in other corners of experimental research on language, multimodality has only recently started featuring more prominently in experimental pragmatic research. A major question which has started receiving more experimental attention relates to how people draw on both vocal and visual information to triangulate intended meaning in face-to-face communication. For instance, work on demonstrative reference has started addressing the extent to which language users are sensitive to pragmatic considerations both when producing and interpreting multimodal utterances containing manual gestures. It has been shown that people modulate the kinematics of deictic pointing gestures when producing multimodal referential expressions in more

or less informative contexts (Peeters et al., 2015a), namely they prolong the duration of the gesture stroke and its poststroke hold phase in order to be more informative. The psychological proximity between a speaker and a pointed referent, as opposed to their actual physical proximity, has also been shown to modulate online ERP signatures in the comprehension of multimodal referring expressions (Peeters et al., 2015b). Finally, more recent work in this line of research has demonstrated that language users have biases in the production of multimodal referring expressions containing verbal and gestural deictics which reflect fine-grained sensitivities to both spatial and speaker-oriented information (Rubio-Fernandez, Long, & Ozyurek, 2026).

Other experimental work in multimodal pragmatics has looked at the impact of manual gestures on quantifier comprehension. Here it has been shown that when interpreting multimodal utterances containing a quantifier and a size-related gesture people modulate their numerical estimates of the respective quantity depending on the form of the gesture (Lorson et al., 2025). Yet another strand of research has looked at the multimodal encoding of information structure. This work has shown that people draw on both prosodic and visual signals when producing utterances with different types of focus (Gregori et al., 2024), and that the two modalities complement one another in how prominently the multimodal utterances express an increase in the pragmatic strength of the focus. This means that while speakers employ various types of pitch accents across focus types, their use of manual gestures and head movements generally increases with an increase in pragmatic strength. All in all, experimental work on the integration between speech and visible gesture points to the complex relationship between vocal and gestural information within an utterance, and to how various pragmatically-relevant sources of information impact both the production and the comprehension of these multimodal utterances in context.

Finally, it's important to note that work in multimodal pragmatics is not restricted to questions of speech-gesture integration. The use of emojis in text-based communication, for instance, also serves as an example of multimodal communication where written language and visual-pictorial information are combined in pragmatically relevant ways. Experimental work in this domain has looked at the interpretation and processing of utterances containing emojis and it has shown, among other things, that emojis are harder to process than meaning-equivalent lexical items

(Tang et al., 2020), that they impact the incremental derivation of irony (Weissman & Tanner, 2018), and that they impact the attribution of speaker commitment (Weissman, 2022, 2024). All these questions relate to core research being done in experimental pragmatics, in particular research pertaining to issues at the semantics-pragmatics interface, such as what the meaning status/ contribution of different linguistic items is and what impact they have on interpretation and processing.

3.3 Commitment and argumentative language use

Much like multimodality, the study of commitment and its attribution is well-established in pragmatics and linguistics (see de Brabanter & Dendale, 2008, and Boulat & Maillat, 2017, for overviews), overlapping with a related line of research in discourse analytic approaches focused on argumentative and persuasive usages of language. Both of these strands of research deal, either more or less directly, with the ways in which language users relate to the truth of what they are communicating in an utterance, and with the ways interpreters attribute accountability of what is said, or implied, to an utterer based on their utterance. As such, commitment blends various notions such as truth, accountability, and speaker meaning, cutting across various scholarly traditions and approaches to linguistic meaning and communication.

Work on commitment is prolific in experimental pragmatic research², and it has mostly built on the theoretical and empirical foundation laid out in relevance theoretic pragmatics. Central questions in this line of research relate to the types of meaning(s) relations language users commit to, and to the extent to which interpreters perceive different degrees of explicitly- and implicitly-encoded speaker commitment. Commitment is commonly measured using tests of deniability or speaker accountability, meaning that participants are asked whether they believe an utterer could deny something they have said, or whether they would trust the utterer's statements in the future after discovering they have communicated false information. Experimental work has shown, for instance, that language users are perceived to be less committed to implicated false content, that is, false content that is derived via an implicature, as opposed to presupposed or asserted false content (Mazzarella et al., 2018). In contrast, users are perceived to be no more committed to explicitly

²See Schumann & Oswald (2026) for a discussion of related experimental work focused on argumentation.

communicated promises compared to promises where the relevant content is derived via a pragmatic enrichment of the linguistically encoded content (Bonalumi et al, 2020). Other experimental findings suggest that users are held more accountable for false information they explicitly communicate as opposed to false information which they communicate implicitly, there being further differences between types of explicitly communicated content (Hall & Mazzearella, 2023). Interestingly, there is considerable variation in the extent to which different types of implicatures are interpreted as lies (Weissman & Terkourafi, 2018), and while scalar implicatures are treated as lies in the strict sense, relevance-based ones are treated as cases of deception where the utterer is not explicitly attempting to convey false information as true (Thalmann et al., 2021). Finally, it has also been shown that false information is detected faster and more accurately when it is expressed as the main point of an utterance, false assertions being identified faster than false presuppositions, the latter also being identified less accurately when it is expressed as background information (Giunta, Mazzearella, & Domaneschi, 2024).

Related experimental work has looked at the strategic use of quantity expressions in argumentative communication. For instance, it has been shown that people use exact numerals in ways that are compatible with an implicature (e.g. “Three [but not four] of the objects in the picture are green”) at a rate similar to plain assertions containing non-vague quantifiers (e.g. “All of the objects in the picture are green”), regardless of whether communicating with cooperative or uncooperative interlocutors (Franke, Dulcinati, Pouscoulous, 2020). In a similar vein, people are more likely to use informationally weaker quantifiers (such as *some* and *most* as opposed to *all*) when they find it difficult to objectively argue for a particular state of affairs (Macuch-Silva et al., 2024). Finally, even quantifiers which denote similar quantities (namely *several* and *a few*) can be perceived as differently strong cues in support of a conclusion a user may be arguing for (Katzir & Ariel, 2024), for instance when “The remote control has several useless buttons” is taken to mean that someone is more dissatisfied with a product compared to someone who states that “The remote has a few useless buttons”.

Put together, experimental work on commitment attribution and perception and the strategic use of vague and underspecified language shows that people use a range of

different pragmatic strategies to achieve their more or less cooperative communicative purposes, and that interpreters are able to pick up on relatively subtle pragmatic cues when making sense of potentially misleading utterances in context. Crucially, this research extends classic experimental pragmatic research by asking in what ways different pragmatic strategies actually impact the extent to which language users are held accountable for what they say, and how these strategies may be used to avoid overtly lying.

4 Conclusion

The present chapter has presented a snapshot of experimental pragmatics. Emerging from experimental psychological work focused on reasoning and linguistically-based inferencing, experimental pragmatics has evolved to become a self-standing field with theoretical interests centered around classic topics in pragmatics, such as reference and speaker meaning, as well as various other questions at the interface between (psycho)linguistics and cognitive science.

As discussed in Section 1, experimental pragmatic work has contributed to core discussions on topics like implicature derivation and figurative language understanding. In particular, experimental work on scalar implicature derivation and metaphor understanding has advanced key discussions in the literature, paving the way for a more detailed empirical picture of these phenomena and more stringent appraisal and testing of theoretical predictions. Section 2 focused on the empirical toolkit of experimental pragmatics. It illustrated the setup of processing experiments in the domains of scalar implicature derivation and metaphor understanding, and it covered the use of computational modeling as a tool to complement experimentation. As discussed, experimental pragmatics has much to gain by further incorporating other empirical tools such as computational modeling in its research cycle, though it should do so critically and by not losing sight of questions of ecological validity and cognitive plausibility. Finally, Section 3 covered some of the current research trends in the field, both more established ones such as probabilistic pragmatics and speaker commitment, as well as relatively underrepresented and emerging ones such as multimodal pragmatics and argumentative language use.

Ultimately, the chapter paints a picture of a field in constant flux, one which adapts to new questions and research tools at the same time as it tackles some of the more fundamental issues in linguistics and the behavioral sciences. Future work in the field should, thus, strive to continue this tradition, so that experimental pragmatics remains a relevant and exciting approach to the study of human language and cognition.

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